

Ranit Mukherjee

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DoB: 15th August, 1992

Education

- 2016–2021 **PhD, Engineering Mechanics, Virginia Tech**, Virginia, USA
- Advisor: Prof. Jonathan B. Boreyko
 - Dissertation title: Exploiting Interfacial Phenomena to Expel Matter from its Substrate
- 2010–2014 **B.Engg., Mechanical Engineering, Jadavpur University**, Kolkata, India

Research Interest Area

Experimental soft matter (complex fluids), Interfacial phenomena, Granular materials, Phase-change heat transfer

Teaching Interest Area

Fluid Mechanics, Heat & Mass Transfer, Thermodynamics, Transport Processes

Teaching Experience

- Spring 2017 **ESM 2304, Introduction to Dynamics, Virginia Tech**, Blacksburg, Virginia
Instructor: Prof. Scott Hendricks and Dr. Jared Gregg
- Fall 2016, **ESM 2104, Introduction to Statics, Virginia Tech**, Blacksburg, Virginia
Fall 2020 Instructor: Prof. Scott Hendricks, Dr. Sneha Davison

Mentoring Experience

- 2017–2018 **Ziad Rashed, Virginia Tech**, Blacksburg, Virginia, USA
Undergraduate researcher, joined a prestigious PhD position at Georgia Tech, USA
- 2023–2025 **Parham Poureslami, University of Minnesota**, Minneapolis, Minnesota, USA
International graduate student, now in his third year of PhD program, already completed a first-author manuscript and co-authored a Soft Matter paper

Professional Experience

- 2021–2025 **Postdoctoral Research Associate, University of Minnesota, Minneapolis**, Minnesota, USA, With Prof. Sungyon Lee
- Led a multi-departmental National Science Foundation (NSF)-sponsored project on engineering the composite behavior of particle-laden interfaces.
 - Mentoring a graduate student in the experimental design and analysis of particle-laden viscous fingering phenomena.

2016–2021 **Graduate Research Assistant**, *Virginia Tech*, Blacksburg, USA, With Prof. Jonathan Boreyko

- Worked in a team to design a novel phase-change thermal-diode prototype for space-sensitive thermal management solutions and characterize the unit's thermal performance.
- Collaborated with a packaging company (formerly Bemis Company, Inc., now Amcor) to develop a low-cost and effective ultra-slippery food-grade packaging solution.
- Lead investigator on pioneering research into spontaneous ice electrification, uncovering jumping frost dendrites as a novel de-icing mechanism. Developed proof-of-concept leading to a collaboration with Rolls-Royce Research and a 500K USD NSF GOALI grant.

2014–2016 **Process Engineer**, *Thermax Limited*, Pune, Maharashtra, India

- Gained practical design experience with direct-fired heaters for processing industries and petrochemical plants, including performing flow-thermal analysis using proprietary software (e.g., FRNC-5PC) and custom in-house VBA code, which I helped develop.
- Worked in a team of process & systems engineers, computer-aided designers, and salespersons to procure million-dollar projects and designed, implemented, and executed them according to customer needs and specifications.

Publications

Journal Publications

(* denotes mentored undergraduate or graduate researcher)

- 2025 P. Poureslami*, **R. Mukherjee** and S. Lee, “*Pattern formation in rectilinear flows of noncolloidal suspensions*”, **Physical Review Fluids**, 11, 064303, 2026
- 2025 B. C. Druecke, A. Hooshanginejad, **R. Mukherjee**, P. Poureslami* and S. Lee, “*Particle-laden filaments from a draining suspension*”, **Soft Matter**, 21, 8479-8488, 2025
- 2025 **R. Mukherjee**, Z. Chen, X. Cheng, and S. Lee, “*Microscopic contact line dynamics dictate the emergent behaviors of particle rafts*”, **Phys. Rev. Fluids**, 10, 084003 , 2025
- 2025 M. Edalatpour, **R. Mukherjee**, and J. B. Boreyko, “*Bridging-Droplet Thermal Diodes: Modeling and Optimization*”, **Int. J. Mass Heat Trans.**, 239, 126594, 2025
- 2023 B. C. Druecke, **R. Mukherjee**, X. Cheng, S. Lee, “*Collapse of a granular raft: transition from single particle falling to collective creasing*”, **Phys. Rev. Fluids**, 8, 024003 , 2023

- 2022 G. J. Iliff*, **R. Mukherjee**, H. A. Gruszewski, D. G. Schmale III, S. Jung, and J. B. Boreyko, “*Phase-change-mediated transport and agglomeration of fungal spores on wheat awns*”, **Journal of Royal Society Interface**, 19, 20210872, 2022
- 2021 **R. Mukherjee**, S.F. Ahmadi, H. Zhang, R. Qiao, and J. B. Boreyko, “*Electrostatic Jumping of Frost*”, **ACS Nano**, 15, 4669–4677, 2021
- 2021 **R. Mukherjee**, H. A. Gruszewski, L. T. Bilyeu*, D. G. Schmale III, and J. B. Boreyko, “*Synergistic dispersal of plant pathogen spores by jumping-droplet condensation and wind*”, **Proc. Natl. Acad. Sci. U.S.A. (PNAS)**, 118, e2106938118, 2021
- 2021 H. Zhang, J. D. Poorter, **R. Mukherjee**, J. B. Boreyko, and R. Qiao, “*Thermoelectrics in ice slabs: charge dynamics and thermovoltages*”, **Phys. Chem. Chem. Phys.**, 23, 16277-16288, 2021
- 2020 M. Edalatpour, K. R. Murphy, **R. Mukherjee**, and J. B. Boreyko, “*Bridging-droplet thermal diodes*”, **Advanced Functional Materials**, 30, 2004451, 2020
- 2019 **R. Mukherjee**, A. S. Berrier*, K. R. Murphy, J. R. Vieitez*, and J. B. Boreyko, “*How surface orientation affects jumping-droplet condensation*”, **Joule**, 3, 1360-1376, 2019
- 2018 **R. Mukherjee**, M. Habibi, Z. T. Rashed*, O. Berbert, X. Shi, and J. B. Boreyko, “*Oil-Impregnated Hydrocarbon-Based Polymer Films*”, **Scientific Reports**, 8, 11698, 2018

Active Patent

- 2026 **US Patent: US12618618B2**, 2023/08/10, “Planar Bridging-droplet thermal diodes, Inventors: J. B. Boreyko, M. Edalatpour, K. R. Murphy, **R. Mukherjee**

In Review or preparation for journal

- 2025 Y. Lolla, V. Kulkarni, **R. Mukherjee**, and J. B. Boreyko, “*Drop Impact on a Lubricant-Infused Fiber*” (In preparation)

Conference Presentations

- 2025 “**Can we predict the death of a granular raft?**”, *Annual Meeting of the American Physical Society Division of Fluid Dynamics*, Salt Lake City, UT, November 23–25 (Oral)
- 2025 “**Elastic or granular: How a continuum model explains the dual nature of granular rafts**”, *Annual Meeting of the American Physical Society Division of Fluid Dynamics*, Salt Lake City, UT, November 23–25 (Oral)
- 2024 “**Collapse of a granular raft: particle-scale features on a continuum model**”, *APS March Meeting 2024*, Minneapolis, MN, March 3–8 (Oral)

- 2023 **“Understanding the collapse of a granular raft”**, *SES Annual Technical Meeting 2023*, Minneapolis, MN, Oct 8–10 (Oral)
- 2023 **“The collapse of a granular raft under bi-axial compression”**, *APS March Meeting 2023*, Las Vegas, NV, March 5–10 (Oral)
- 2022 **“On the collapse of a granular raft in a funnel”**, *Granular Matter Gordon Research Conference*, Stonehill College, MA, USA, June 27–July 1, 2022 (Poster)
- 2021 **“Student Keynote Award Presentation: Jumping Ice”**, *Inaugural micro Flow and Interfacial Phenomena Conference*, Virtual, June 7–9, 2021 (Oral)
- 2019 **“Jumping Frost”**, *72nd Annual Meeting of the American Physical Society Division of Fluid Dynamics*, Seattle, WA, November 23–26, 2019 (Oral)
- 2019 **“How Surface Orientation Affects Jumping-Droplet Condensation”**, *Gordon Research Conference on Micro and Nanoscale Phase Change Heat Transfer*, Lucca, Italy, February 3–8, 2019 (Poster)
- 2018 **“Oil-infused Polyethylene Films”**, *MII Technical Conference and Review*, Virginia Tech, Blacksburg, April 16–18, 2018 (Poster)
- 2018 **“Effect of Surface Orientation on Jumping-droplet Condensation”**, *16th International Heat Transfer Conference (IHTC-16)*, Beijing, China, August 10–15, 2018 (Poster)
- 2017 **“Oil-infused Polyethylene Films”**, *Annual Meeting of the American Physical Society Division of Fluid Dynamics*, Denver, CO, November 19–21, 2017 (Oral)

Outreach Programs as a Science Educator

- 2024 **Squishy Science Sunday**, *This was an outreach event organized to introduce concepts of soft matter physics to a general audience*, APS March Meeting, Minneapolis, Minnesota
- 2017–2020 **Virginia Tech Science Festival**, *Yearly expo-style, family-friendly events to engage with graduate scientists. Festival guests take part in hands-on activities and demonstrations at about 100 different exhibits*, Blacksburg, Virginia
- 2017–19 **C-Tech² Summer Camp**, *Yearly summer camp activity aimed at rising junior and senior high school girls. The purpose is to provide access to information for a successful STEM career. Organized by the Center for the Enhancement of Engineering Diversity (CEED)*, Blacksburg, Virginia
- 2017, 2018 **Kids Tech University (KTU)**, *An educational outreach program to inspire children between ages 9–12 years in STEM education*, Blacksburg, Virginia

Professional Honor

- 2026 **Editor’s Suggestion for a publication**, *Physical Review Fluids*
- 2021 **Student Keynote Presentation Award**, *μ FIP Conference*
- 2021 **My work on jumping frost was reported on the popular science website FYFD**
- 2019 **Daniel and Frances Frederick Scholarship**, *Virginia Tech, USA*
- 2018 **Top 100 in Materials Science among all the papers published in**, *Scientific Reports*

2018 **Liviu Librescu Memorial Fellowship**, *Virginia Tech, USA*

Professional Activities

- 2016-2021 **Peer review: Co-reviewing with advisor**, *Scientific Reports, Nano Energy, Advanced Functional Materials, Physical Review Letters, ACS Nano, Soft Matter, ACS AMI, Langmuir, Physical Review Fluids*
- 2021-present **Peer review: Independent**, *Advanced Science, Newton, Int. J. Heat & Mass Transfer*
- 2016-2021 **Past Member**, *Bio-Inspired Science & Technology Center at Virginia Tech*
- 2016-2021 **Past Member**, *Macromolecules and Interfaces Institute at Virginia Tech*

References

- Sungyon Lee **Associate Professor**, *Postdoc advisor*,
Department of Mechanical Engineering, University of Minnesota-Twin Cities,
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- Xiang Cheng **Professor**, *Collaborator*,
Department of Chemical Engineering and Materials Science, University of
Minnesota-Twin Cities, Minneapolis, MN 55455, USA,
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- Jonathan B. Boreyko **Associate Professor**, *PhD advisor*,
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